Ajeenkya DY Patil School of Engineering (Polytechnic), Charholi

Via Lohegaon, Pune-412 105

Department of Computer Engineering

2025-26[6th Sem]

Project Report

Of

Capstone Project

On

"PeerLink: Decentralized Person to Person Messenger"

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Department Of Computer Engineering
2025-26[6th Sem]

CERTIFICATE

This is to certify that Project report entitled **"PeerLink: Decentralized Person to person messenger"** is submitted in the partial fulfillment of requirement for the award of the Diploma in Computer Engineering by Maharashtra State Board of Technical Education as record of student's own work carried out by them under the guidance and supervision at Ajeenkya DY Patil School of Engineering (Charholi), during the academic year 2025- 26.

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ABSTRACT

The contemporary digital communication paradigm is predominantly anchored in centralized infrastructure, relying heavily on cloud-based servers and wide-area network routing. While highly efficient under stable conditions, this topology is inherently fragile, presenting critical single points of failure during infrastructural collapse, such as natural disasters, or in geographically isolated topographies. Furthermore, reliance on third-party servers inherently introduces vulnerabilities regarding data sovereignty and privacy surveillance. To counteract these limitations, this project presents the theoretical framework and pragmatic implementation of PeerLink, an autonomous, infrastructure-free peer-to-peer (P2P) localized messaging ecosystem. PeerLink challenges the dependency on global internet connectivity by advancing the concept of edge-defined networking, wherein a low-power microcontroller (the ESP32) is conceptualized as a localized, portable Access Point (AP) and dynamic datagram router.

Operating completely independently of cellular networks, Internet Service Providers (ISPs), and Subscriber Identity Modules (SIMs), the proposed architecture establishes an isolated, self-sustaining wireless domain. At its conceptual core, the system utilizes a stateless User Datagram Protocol (UDP) multiplexing mechanism to facilitate instantaneous node-to-node communication. The centralized edge-router operates as a dynamic registry, continuously mapping client presence and executing decentralized packet forwarding based on self-discovered IP allocations within the local subnet. By offloading routing logic to a physical layer edge device, the architecture eliminates the necessity for cloud-based packet mediation. Concurrently, the mobile client architecture is modeled on reactive state management and localized data persistence, ensuring seamless, user-agnostic network discovery, foreground lifecycle maintenance, and asynchronous data flow.

Theoretical evaluation of this architecture demonstrates a significant paradigm shift in off-grid communication. By bypassing Transmission Control Protocol (TCP) handshake latency and leveraging stateless short-burst data packets, the network achieves near-zero latency text transmission within its localized physical radius. Moreover, by confining packet traversal strictly within the Local Area Network (LAN) and stripping away external routing dependencies, the system inherently guarantees end-to-end data localization, nullifying the possibility of external server interception. Ultimately, this research substantiates the viability of decentralized edge-routing ecosystems. It establishes a robust foundational architecture for scalable, autonomous networks, presenting a highly resilient communication framework necessary for disaster response, strategic remote deployments, and fully localized, privacy-preserving interactions.

Looking forward, the theoretical expansion of this autonomous domain envisions a multi-hop, highly scalable mesh network architecture. By algorithmically chaining multiple edge-routing nodes, the localized subnet could dynamically expand its geographical footprint across vast topological spaces, overcoming hardware traversal limitations without reverting to centralized internet backbones. Concurrently, the integration of advanced cryptographic primitives—such as localized asymmetric encryption algorithms and decentralized key exchange protocols—would theoretically harden the P2P transport layer. This conceptual trajectory ensures that even in hostile physical environments or under proximity-based surveillance, the self-healing network maintains absolute data integrity, forward secrecy, and unconditional anonymity, thereby fully cementing decentralized edge communications as the future of resilient digital infrastructure.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AP	Access Point
API	Application Programming Interface
ESP32	Espressif Systems 32-bit Microcontroller
IP	Internet Protocol
JSON	JavaScript Object Notation
LAN	Local Area Network
MVVM	Model-View-ViewModel
OS	Operating System
P2P	Peer-to-Peer
SDK	Software Development Kit
SSID	Service Set Identifier
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
UI	User Interface
WLAN	Wireless Local Area Network
Wi-Fi	Wireless Fidelity

CHAPTER 1: INTRODUCTION

1.1 Project Overview

In the contemporary era of the "Internet of Things" (IoT) and hyper-connectivity, the vast majority of our digital interactions are mediated by centralized infrastructures. Whether it is text messaging, voice calls, or data sharing, the fundamental architecture relies on a "Device-to-Server-to-Device" path. While this model is highly efficient for global communication, it introduces a significant vulnerability: the requirement of an active internet gateway.

PeerLink is a specialized project aimed at reclaiming communication autonomy in environments where this centralized infrastructure is absent, compromised, or intentionally restricted. By merging the capabilities of embedded systems (microcontrollers) with modern mobile software (Android), PeerLink establishes a localized, autonomous communication "island."

The system utilizes an ESP32 microcontroller as a dedicated, low-power edge router. This hardware serves as a portable Wi-Fi Access Point (AP) that manages a private subnet. Nearby Android devices, equipped with the PeerLink client application, can detect, authenticate, and connect to this hardware automatically. Once connected, they form a Peer-to-Peer (P2P) network where messages are routed directly through the ESP32. This approach eliminates the need for expensive cellular data, SIM cards, or reliable ISP signals, providing a robust "Off-Grid" messaging solution.

1.2 Need of the Project

The necessity for an offline communication framework like PeerLink is driven by three distinct pillars: Safety, Sovereignty, and Cost.

1.2.1 Critical Safety and Disaster Recovery

During high-magnitude natural disasters such as earthquakes, tsunamis, or hurricanes, the physical integrity of cellular towers and fiber-optic cables is often the first to fail. In these critical moments, "Golden Hour" coordination becomes impossible. PeerLink meets this need by providing a pocket-sized network that can be deployed instantly, allowing victims and emergency responders to communicate locally without needing the grid to be online.

1.2.2 Information Sovereignty and Privacy

In an age of mass data surveillance, many users and organizations require "Zero-Leak" communication for sensitive topics. Since traditional messaging apps route data through third-party servers, they are subject to legal subpoenas or hacking risks. PeerLink ensures that packets never leave the 100-meter radius of the physical hardware, guaranteeing that communication remains truly private and localized.

1.2.3 Remote and Frontier Environments

Outdoor enthusiasts, trekking groups, maritime explorers, and underground mining teams often operate in "Net-Zero" zones. Providing satellite internet to every individual in these groups is economically and logistically unfeasible. PeerLink offers a low-cost, multi-user relay that allows a group of explorers to stay in constant contact within their immediate vicinity for zero recurring cost.

1.3 Problem Statement

The central problem addressed by this project is the "Infrastructural Dependency" of modern communication.
Currently, if two users are standing 10 meters apart but in an area with no cellular reception, they cannot exchange a digital message using standard applications. This "Paradox of Modern Connectivity" means that our most advanced tools are useless without a central authority (the ISP).
There is a critical lack of a user-friendly, high-performance, and affordable platform that can:

1. Automatically bridge the connection gap between mobile devices.
2. Provide real-time messaging without manual IP configuration.
3. Offer persistent data logs for localized history.
4. Operate with extreme power efficiency for long-term deployment on batteries.

1.4 Scope of the Project

The scope of PeerLink is meticulously defined to ensure a high-quality, specialized deliverable:

1.4.1 Functional Scope

- Hardware Routing: Implementation of a stateless UDP routing engine on the ESP32.

- P2P Messaging: Enabling real-time text transmission and presence broadcasting.

1.4.2 Non-functional Scope

The non-functional requirements define the performance, usability, and efficiency standards of the system. These requirements ensure that the system operates reliably under various conditions and provides a seamless user experience.

- Latency:
  The system is designed to achieve a response time of less than 100 milliseconds for local packet delivery. This ensures fast communication between devices, which is critical in time-sensitive or emergency situations.
- Power Optimization:
  The system focuses on efficient power management to enable the ESP32 module to operate continuously for more than 48 hours using a standard 10,000 mAh power bank. Techniques such as sleep modes and optimized processing are utilized to minimize energy consumption.
- Ease of Use:
  The user interface (UI) is designed for simplicity and accessibility, allowing one-handed operation. It ensures high readability, even in high-stress or low-visibility conditions, by using clear layouts, large fonts, and intuitive controls.

1.5 Motivation

The core motivation stems from the realization that “connectivity is a prerequisite for safety.” During research, we identified that many lives are lost during remote expeditions or urban crises simply because victims were unable to send a signal to a group leader or a nearby responder. Furthermore, as developers, we were motivated by the technical challenge of "Zero-Infrastructure" networking—creating something out of nothing. By turning a ₹400 microcontroller into a sophisticated routing hub, we aim to demonstrate that high-impact technology can be both affordable and life-saving.

1.6 Objectives

The project is governed by five primary objectives:
1. Hardware Hub Design: To create a dedicated C++ firmware that transforms the ESP32 into a broadcast listener and router on Port 8888.
2. Discovery Automation: To implement a scanning algorithm in Kotlin that filters SSIDs and establishes a secure Wi-Fi bond without user intervention in settings.
3. Real-Time Data Sync:To utilize the User Datagram Protocol (UDP) for asynchronous "Fire-and-Forget" messaging to minimize power and latency.
4. User Experience (UX) Goal: To build a reactive Material 3 interface that provides instant visual feedback on network status and peer availability.

1.7 Technologies Used

The PeerLink project represents a "Full-Stack Hardware/Software" implementation. The following technologies were selected for their performance and compatibility:

Component Category	Technology Used	Version / Spec	Primary Rationale in PeerLink
Microcontroller	ESP32 DevKit V1	Xtensa Dual-Core	Integrated 2.4GHz Wi-Fi and extremely low power for portable use.
Firmware SDK	Arduino ESP32 Core	v2.0.1 / 3.x	Robust library support for Wi-Fi AP and UDP socket management.
Mobile Platform	Android OS	API 29 (10) - 34	Support for WifiNetworkSpecifier which allows programmatic hotspot bonding.
App Language	Kotlin	v1.9.22	Native Android performance with powerful Coroutines for async networking.
UI Design	Jetpack Compose	BOM 2024.02	Modern, reactive UI that automatically updates when messages arrive.
Local Storage	Room Database	v2.6.1	High-performance persistence over SQLite for locally saved chat logs.
Network Protocol	UDP (Stateless)	Port 8888	Minimal overhead for high-speed delivery without TCP handshakes.
Data Format	JSON	RFC 8259 Standard	Lightweight data exchange format compatible across C++ and Kotlin.
Concurrency	Kotlin Flow	Cold Streams	Reactive data streaming from the DB to the UI in real-time.

Table 1.1: Technology Stack And Versioning Overview

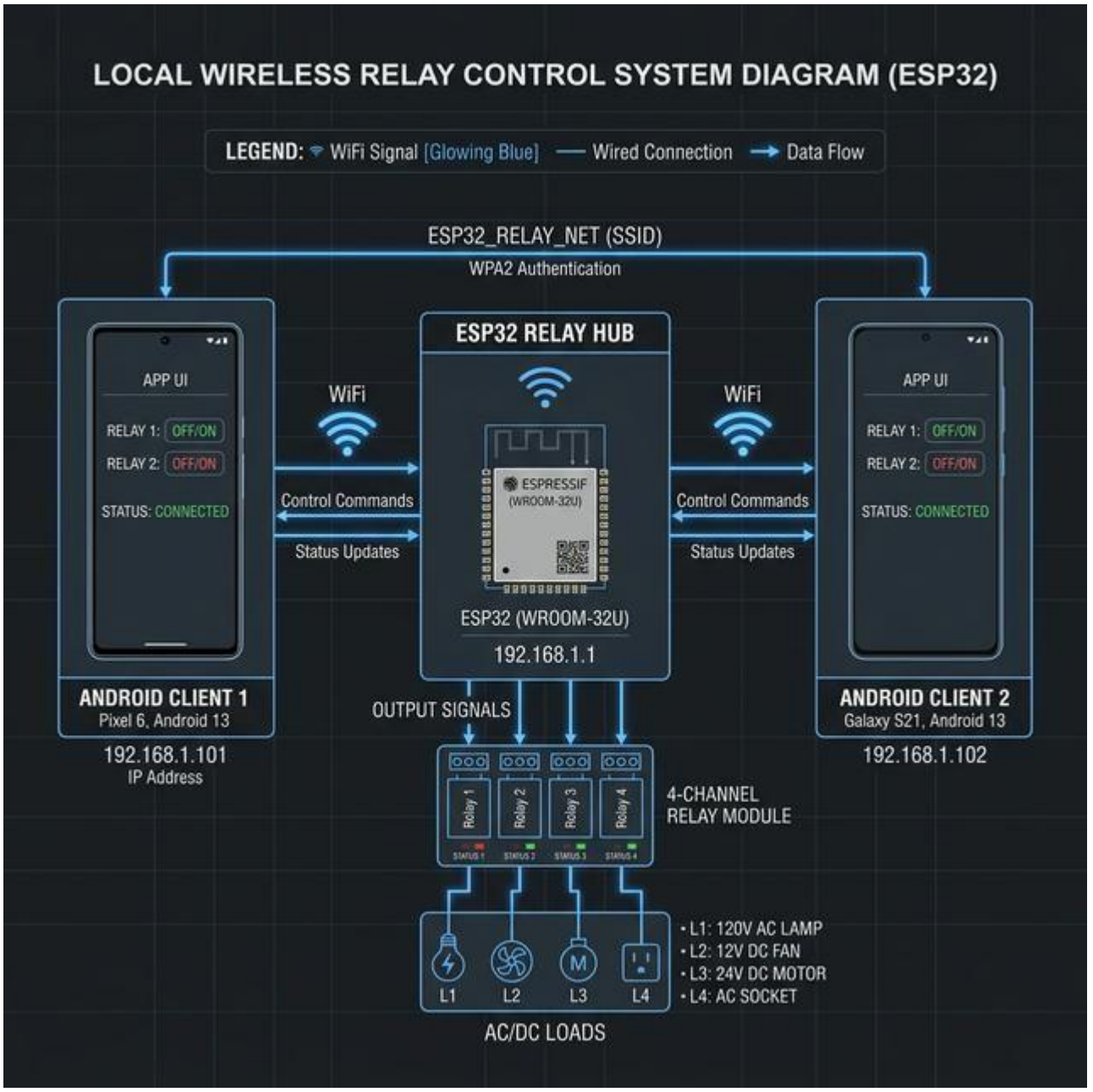

Figure 1.1: PeerLink Conceptual Communication Model

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The quest for decentralized communication has existed since the inception of computer networking. While the modern world is dominated by client-server architectures, the theoretical foundations of Peer-to-Peer (P2P) systems prove that resilience is best achieved at the edges of the network. This survey explores the various technologies and existing implementations that attempt to provide offline or localized messaging services.

2.2 Study of P2P Communication

Peer-to-Peer (P2P) communication is a decentralized model where each participant in the network acts as both a client and a server. In the context of mobile devices, P2P communication eliminates the need for a central administrative server, allowing devices to exchange data directly.
- Advantages: High resilience, no single point of failure, and scalability.
- Challenges in Mobile: Energy consumption, dynamic topography (nodes moving in and out of range), and IP address management.

2.3 Study of Wi-Fi Direct Technology

Wi-Fi Direct is a standard that allows devices to connect to each other without requiring a wireless access point.
- Mechanism: It uses a "Group Owner" (GO) negotiation process where one mobile device acts as an AP for others.
- Limitations: The connection process is often cumbersome, involving manual discovery and "Invite/Accept" prompts that interrupt the user experience. Furthermore, older Android versions and different hardware manufacturers have inconsistent implementations of the Wi-Fi Direct stack.

2.4 Study of Offline Messaging Applications

Several applications have attempted to solve the offline messaging problem using various protocols:
1. Bridgefy: Uses Bluetooth Mesh networking. While effective, Bluetooth has a very limited range (10-15 meters) and lower data throughput compared to Wi-Fi.
2. FireChat (Discontinued): One of the pioneers in multi-hop mesh networking for protesters and crisis zones. It relied heavily on Open Garden's proprietary mesh technology.
3. Briar: An open-source app focusing on high security and synchronization over Bluetooth, Wi-Fi, and Tor. It is highly resilient but has a steep learning curve.

2.5 Study of Similar Application – Bitchat

Bitchat is a well-known project in the open-source community that utilizes Bluetooth and Wi-Fi Direct for localized communication.
- Strengths: Lightweight and focused on privacy.
-Weaknesses: It relies on the mobile devices themselves to handle both routing and communication. This leads to high battery drain as the phone's radio must constantly stay in an active scanning/broadcasting state. PeerLink improves upon this by offloading the "Server/Router" role to an external, low-power ESP32 device.

2.6 Limitations Found in Previous Systems

Based on the survey, several recurring limitations were identified:
1. Hardware Dependency: Many apps work on some phones but not others due to fragmented Wi-Fi Direct support.
2. Battery Drain: Acting as a "Hotspot" or "Mesh Node" severely depletes a smartphone's battery life.
3. Range: Bluetooth-based systems are inadequate for large-scale coordinate efforts in disasters.
4. Complex UX: Setting up a mesh connection often requires technical knowledge beyond the average user.

2.7 Chapter Summary

Feature	**Bluetooth Mesh (Bridgefy)**	**Wi-Fi Direct (Bitchat)**	**PeerLink (Proposed System)**
Typical Range	10 – 20 meters	30 – 50 meters	70 – 100+ meters
Data Throughput	Very Low	High	Medium to High
Battery Impact	Medium	High	Low (Processing offloaded to ESP32)
Ease of Use	Moderate	Complex	Seamless (Automatic Connection)
Infrastructure	Not Required	Ad-hoc Network	Portable Edge Node-Based System

Table 2.1: Comparative Analysis of Existing Offline Messaging Solutions

The literature survey confirms that while P2P messaging is conceptually sound, the practical implementation on smartphones suffers from hardware fragmentation and battery issues. PeerLink addresses these by introducing a dedicated hardware relay layer, combining the high speed of Wi-Fi with the reliability of a dedicated router.

CHAPTER 3: SCOPE OF THE PROJECT

3.1 Introduction

The scope of a project defines the boundaries, deliverables, and requirements necessary to achieve a successful implementation. For PeerLink, the scope extends from the low-level embedded programming of the ESP32 to the high-level reactive UI development of the Android application. This chapter analyzes the current state of communication and details the requirements for our proposed offline solution.

3.2 Existing System Analysis

The current communication ecosystem is built on a "Centralized Spoke-and-Hub" model.
- Dependency: Users are dependent on a global internet gateway.
- Failure Mode: If the internet provider or a local cell tower fails, the entire application layer becomes useless.
- Privacy: Data is stored on remote servers, making it susceptible to data breaches and surveillance.
- Cost: equires active data plans or paid Wi-Fi access.

3.3 Requirement Analysis

To build a viable alternative, the system must fulfill the following requirements:
1. Autonomous Networking: Must create a network from scratch without external help.
2. Zero-Configuration: Users should be able to join the network with minimal technical steps.
3. Low Latency: Messaging must be real-time to be effective during emergencies.
4. Persistent Storage: Messages must be saved locally on the device for later reference.
5. Reliability: The ESP32 relay must handle multiple concurrent connections without crashing.

3.4 Proposed System

The proposed PeerLink system introduces an "Edge-Relay" architecture.
- The Gateway (ESP32): Acts as a hardware router that manages client IP addresses and routes UDP packets based on their destination ID.
- The Client (Android): A custom app that intercepts the WiFi state, connects to the "PeerLink" SSID, and provides a messaging interface.
- Messaging Protocol: Uses JSON-encoded packets over UDP port 8888 for speed and simplicity.

3.5 Feasibility Analysis

- Technical Feasibility: ESP32 supports AP mode and UDP protocols natively. Android provides the `WifiNetworkSpecifier` API for programmatic connections.
- Economic Feasibility: The hardware cost is less than ₹400, making it extremely affordable for mass deployment.
- Operational Feasibility: The app is designed for simple, intuitive use cases, requiring no special training for general users.

3.6 Chapter Summary

PeerLink is a feasible and necessary project that bridges the gap between total connectivity and total isolation. By delegating the heavy lifting of networking to an external microcontroller, it creates a sustainable and high-performance offline environment.

3.7 Hardware and Software Requirements

Specification	Details
Microcontroller	Xtensa Dual-Core 32-bit LX6
Clock Frequency	Up to 240 MHz
Operating Voltage	3.3V (Core) / 5V (via USB)
SRAM / Flash	520 KB SRAM / 4 MB Flash
Wi-Fi	IEEE 802.11 b/g/n (up to 150 Mbps)
Antenna	Built-in PCB Trace Antenna
Power Source	Power Bank / Mobile Charger / External Battery

Table 3.1: Hardware Component Specifications (ESP32 DevKit V1)

Component	Minimum Requirement
Operating System	Android 10 (API Level 29) or Higher
RAM	Minimum 2 GB
Connectivity	Wi-Fi (2.4 GHz support required)
Permissions	Fine Location, Wi-Fi State Access, Notifications
SDK Libraries	Kotlin, Jetpack Compose, Room, Gson
Storage	10 MB (App) + Additional Space for Database

Table 3.2: Minimum Software Requirements for Android Client

CHAPTER 4: METHODOLOGY

4.1 Introduction

The methodology of the PeerLink project follows an iterative development life cycle, focusing on the seamless integration of embedded firmware and mobile application logic. This chapter details the technical architecture, operational phases, and design modules that make the offline peer-to-peer communication possible.

4.2 System Architecture

The architecture is designed as a Star Topology where the ESP32 acts as the central hub (Relay Node) and various Android phones act as leaf nodes.

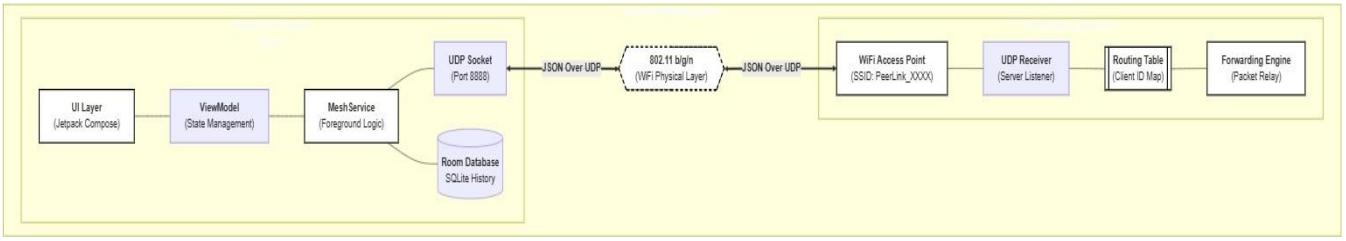

Figure 4.1: High-Level System Architecture Diagram

4.3 Working of the System

The system operates in three distinct phases which ensure a structured and reliable connection lifecycle:

1. Discovery Phase: The ESP32 broadcasts its presence via beacon packets. The Android app uses the `WifiNetworkSpecifier` API to scan for these beacons and automatically establish a secure hardware bond with the `PeerLink_XXXX` SSID.

2. Registration Phase: Upon successful connection, the Android app sends a `presence` packet containing its unique Device ID and friendly name. The ESP32 receives this data and persists it in its internal `ClientInfo` map (Routing Table).

3. Communication Phase: When User A sends a message to User B, the app encapsulates the text into a JSON packet directed to User B's ID. The ESP32 performs a map-lookup to find User B's IP address and instantly forwards the datagram.

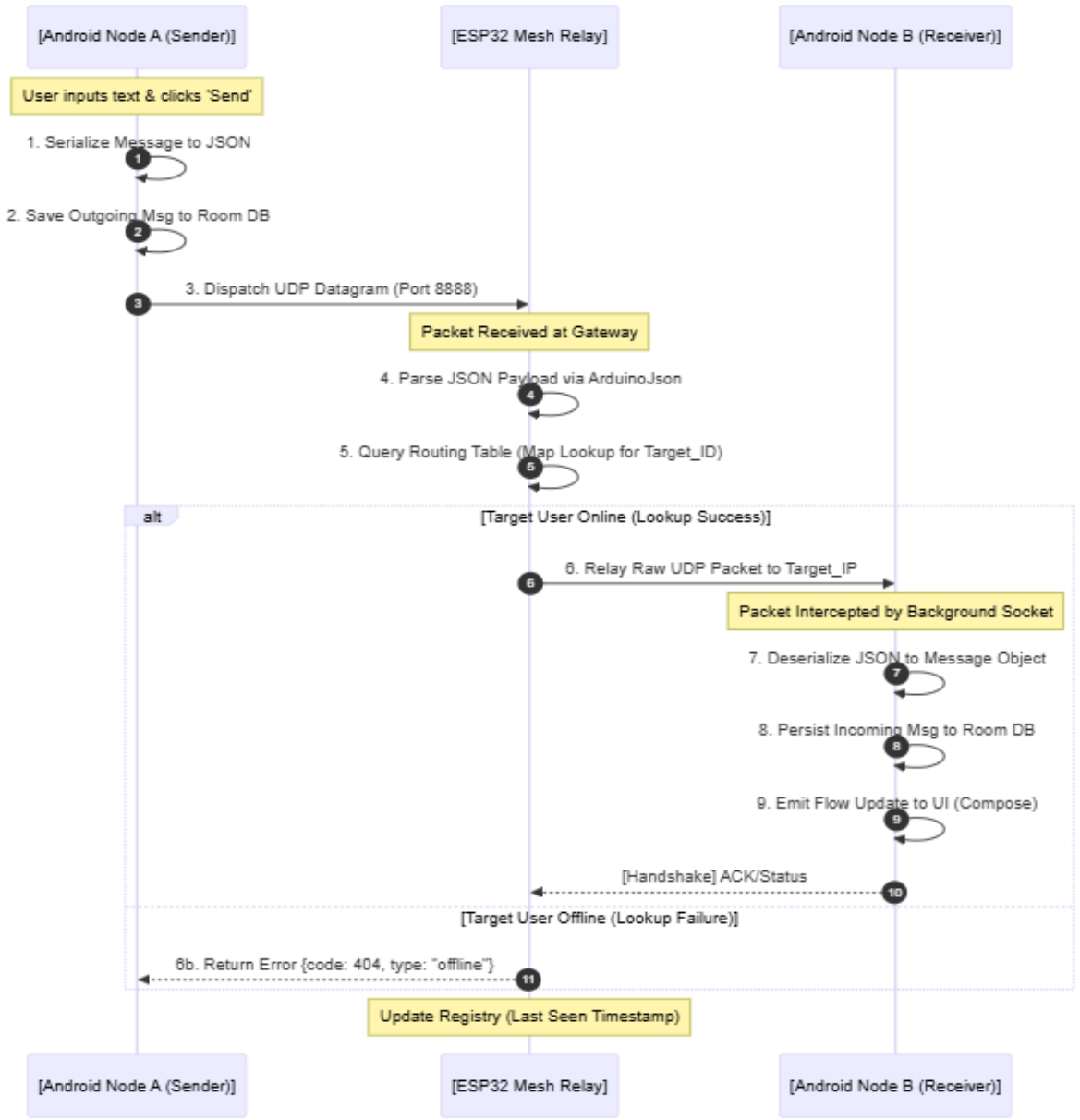

Figure 4.2: Real-time Communication Sequence

4.4 Module Design

4.4.1 ESP32 Relay Module

The firmware is responsible for the low-level infrastructure of the network.
- AP Management: Maintaining the Wi-Fi Access Point stability.
- Dynamic Routing: Managing a volatile memory map of connected client IDs and their dynamic IP addresses.
- Garbage Collection: An automated 30-second timeout mechanism that removes inactive clients to free up memory.

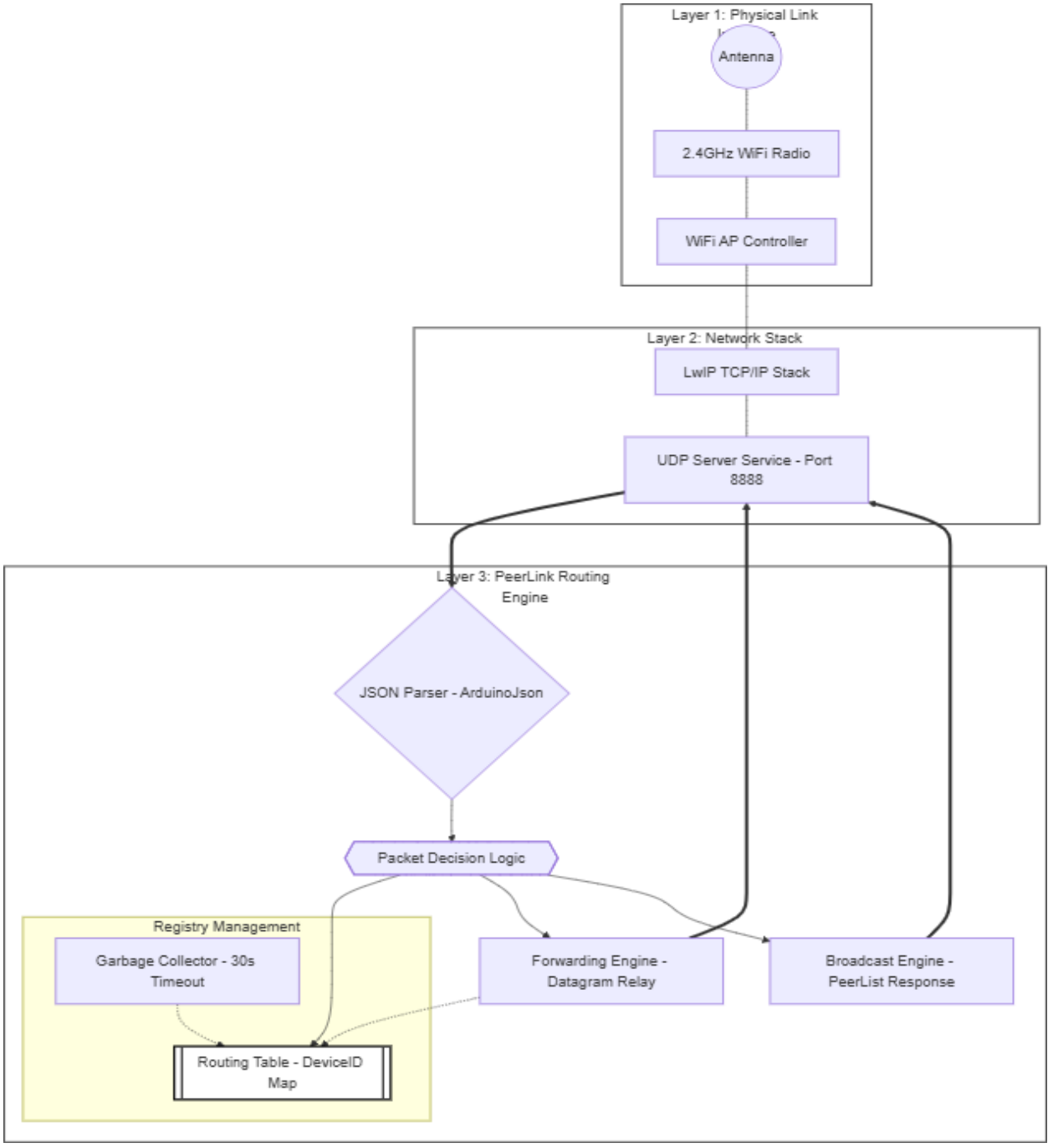

Figure 4.3: ESP32 Edge-Router Block Diagram

4.4.2 Android Networking Module (UdpChat)

This module handles the transmission and reception of raw packets. It utilizes a `DatagramSocket` bound to port 8888. It runs within a high-priority background thread to ensure no packets are dropped during heavy traffic.

4.4.3 Android Persistence Module (Room DB)

To ensure communication continuity, every message—whether sent or received—is stored in a local SQL database. This ensures that the chat history remains persistent even if the hardware connection is lost or the application is restarted.

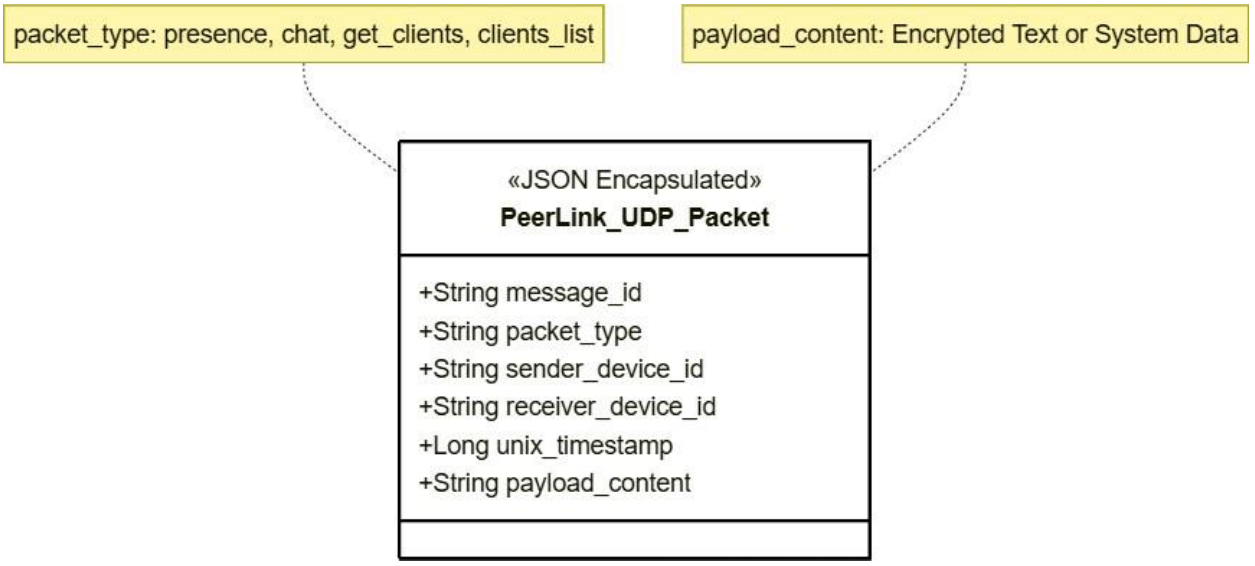

Figure 4.4: UDP Packet Structure (JSON Encapsulation)

4.5 Data Flow

The data flow within the PeerLink ecosystem is represented by the movement of encapsulated JSON packets.

1. Input: User types a message and clicks 'Send'.
2. Process 1: App encapsulates the message into a JSON object and sends it over UDP.
3. Process 2 (Router): The ESP32 receives the packet, parses the JSON, and identifies the target recipient.
4. Process 3: The ESP32 forwards the packet to the target recipient's specific IP address.
5. Output: The receiver app decodes the JSON and reactively displays the message.

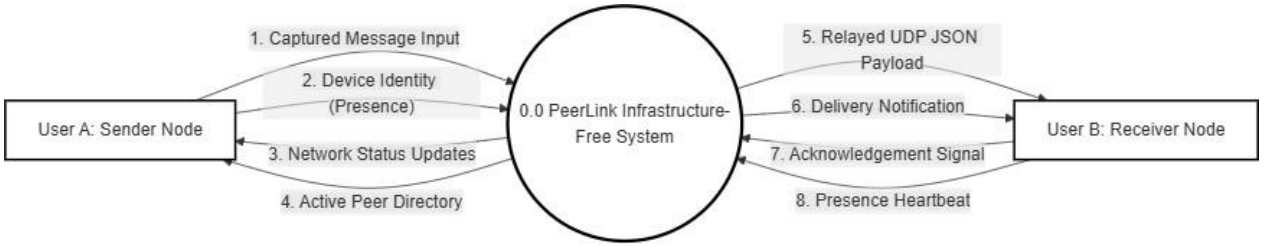

Figure 4.5: Data Flow Diagram (DFD) - Level 0

Type	Direction	Purpose
presence	Client → ESP	Registers a new device on the network
chat	Client ↔ Client	Transmits actual message content between users
get_clients	Client → ESP	Requests the list of currently connected (online) users
clients_list	ESP → Client	Returns the compiled list of peer IDs and user names

Table 4.1: UDP Packet Payload Types and Definitions

4.6 Use Case

The system facilitates specific interactions between the users and the hardware environment.
- Sender Use Cases: Connect to PeerLink, View Online Peers, Send Message, View History.
- Receiver Use Cases: Receive Message Notification, Automated Peer Registration, Store Message.
- System Use Cases: Manage Routing Table, Handle Timeouts, Broadcast Presence.

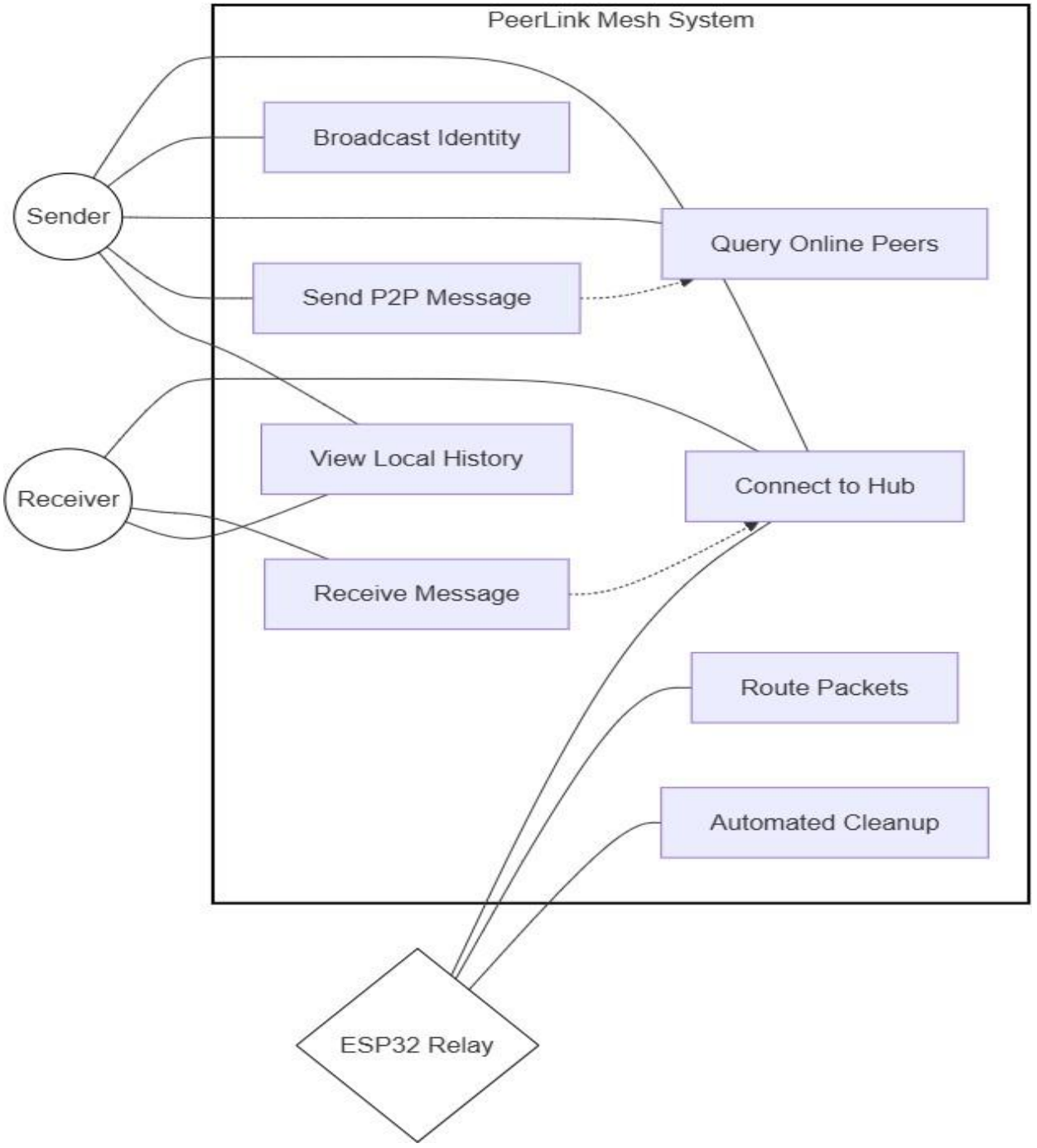

Figure 4.6: PeerLink System Use Case Diagram

4.7 Interface Design

The user interface is designed with a "Dark Mode" Typography to minimize OLED power consumption and ensure high readability in challenging outdoor conditions.

The Splash Screen: Focused on high-impact branding and initializing background services.
Discovery Screen: Features a radial pulsing animation to provide visual feedback during hardware searching.
Device List: A reactive material list showing the friendly names of all currently connected peers.
Chat Interface: A standard bubble-style interface designed for rapid, intuitive messaging with UTC

4.8 Chapter Summary

The methodology focuses on making the complexity of networking invisible to the end-user. By using standardized JSON packets and reliable UDP routing, PeerLink provides a robust framework that mimics the behavior of online messaging apps while remaining entirely on the edge and internet-independent.

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CHAPTER 5: DETAILS OF DESIGN, WORKING AND PROCESSES

5.1 Development Environment

The implementation of PeerLink required a well-structured cross-platform development environment to effectively bridge the gap between embedded hardware and mobile software systems.
The Embedded Layer was developed using Arduino IDE 2.3.2 integrated with the Xtensa ESP32 toolchain. Essential libraries such as *WiFi.h* were utilized for Access Point configuration, while *WiFiUDP.h* enabled efficient socket-level communication for data transmission.
The Mobile Layer was designed and developed using Android Studio (Ladybug) with Kotlin 1.9.22 as the primary programming language. The application build process was managed using Gradle 8.x, and the user interface was implemented using the Jetpack Compose BOM (2024.02) to ensure a modern, reactive, and responsive UI experience.
The Database Layer was implemented using the Room Persistence Library, which provides a robust abstraction over SQLite. This enabled efficient local data storage, ensuring reliable offline data logging and retrieval.
The Version Control System was managed using Git, allowing systematic tracking of changes, collaboration, and synchronization between firmware and application development throughout the project lifecycle.

5.2 Module Implementation

5.2.1 ESP32 Module (Back-end Mesh Controller)

The ESP32 firmware is designed as a Stateless Packet Relay, responsible for managing communication between connected devices without maintaining persistent storage. It creates a private SSID network and maintains a temporary *Peer Map* in its volatile memory to track active devices within the network. The module incorporates an AP Manager, which configures the ESP32 in Access Point mode, manages WPA2 encryption for secure communication, and handles DHCP distribution to dynamically assign IP addresses to connected clients. Additionally, a UDP Dispatcher operates as a continuous listener on Port 8888, receiving incoming data packets and processing JSON-encapsulated messages to perform operations such as device registration, message forwarding, and broadcasting of the client list.

5.2.2 Android Client Module (Front-end Mesh Interface)

The Android client application is designed using a Modular MVVM (Model-View-ViewModel) Architecture, ensuring a clear separation of concerns, improved scalability, and maintainability. The system includes a Network Interface (UdpChat), which manages communication with the ESP32 using UDP sockets and utilizes Kotlin Coroutines to perform non-blocking input/output operations, thereby maintaining smooth and responsive performance. In addition, a Foreground Service (MeshService) runs continuously in the background to maintain an active connection with the mesh network, ensuring uninterrupted communication even when the device screen is off or the application is running in the background.

5.3 Hardware Implementation

5.3.1 Hardware Setup: ESP32 DevKit V1 and Connectivity

The hardware implementation of the PeerLink system is designed with a focus on portability, simplicity, and efficiency. The setup primarily consists of the ESP32 DevKit V1 module, which serves as the central communication controller. The ESP32 is powered using a standard 5V USB source, such as a power bank or mobile charger, making the system highly portable and suitable for deployment in remote or emergency environments.
Internally, the supplied 5V power is regulated to 3.3V using an onboard voltage regulator, which is required for the proper operation of the ESP32 microcontroller and its associated components. The integrated 2.4 GHz Wi-Fi module enables wireless communication, while the built-in PCB antenna ensures reliable signal transmission and reception. Additionally, a status LED connected to a GPIO pin provides visual feedback regarding system activity, such as device connection or data transmission.
This minimal hardware configuration eliminates the need for additional external components, thereby reducing system complexity, cost, and power consumption while maintaining efficient and reliable performance.

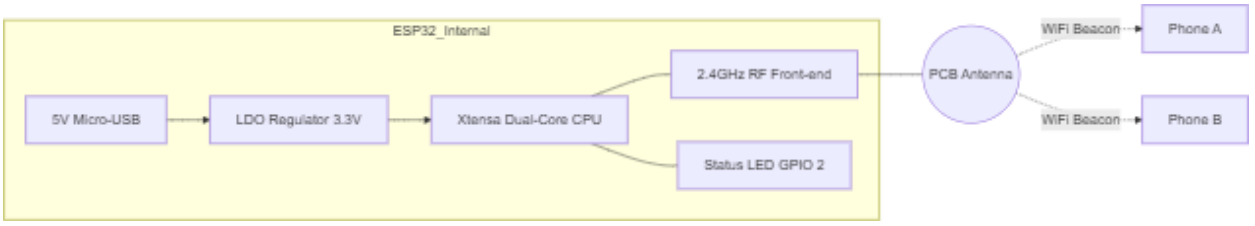

Figure 5.1: Hardware Setup: ESP32 DevKit V1 and Connectivity

5.3.2 Firmware Execution Logic (Flowchart)

The firmware follows a Reactive Polling Pattern. It rests in a low-power listening state and triggers routing logic only upon the arrival of valid UDP packets.

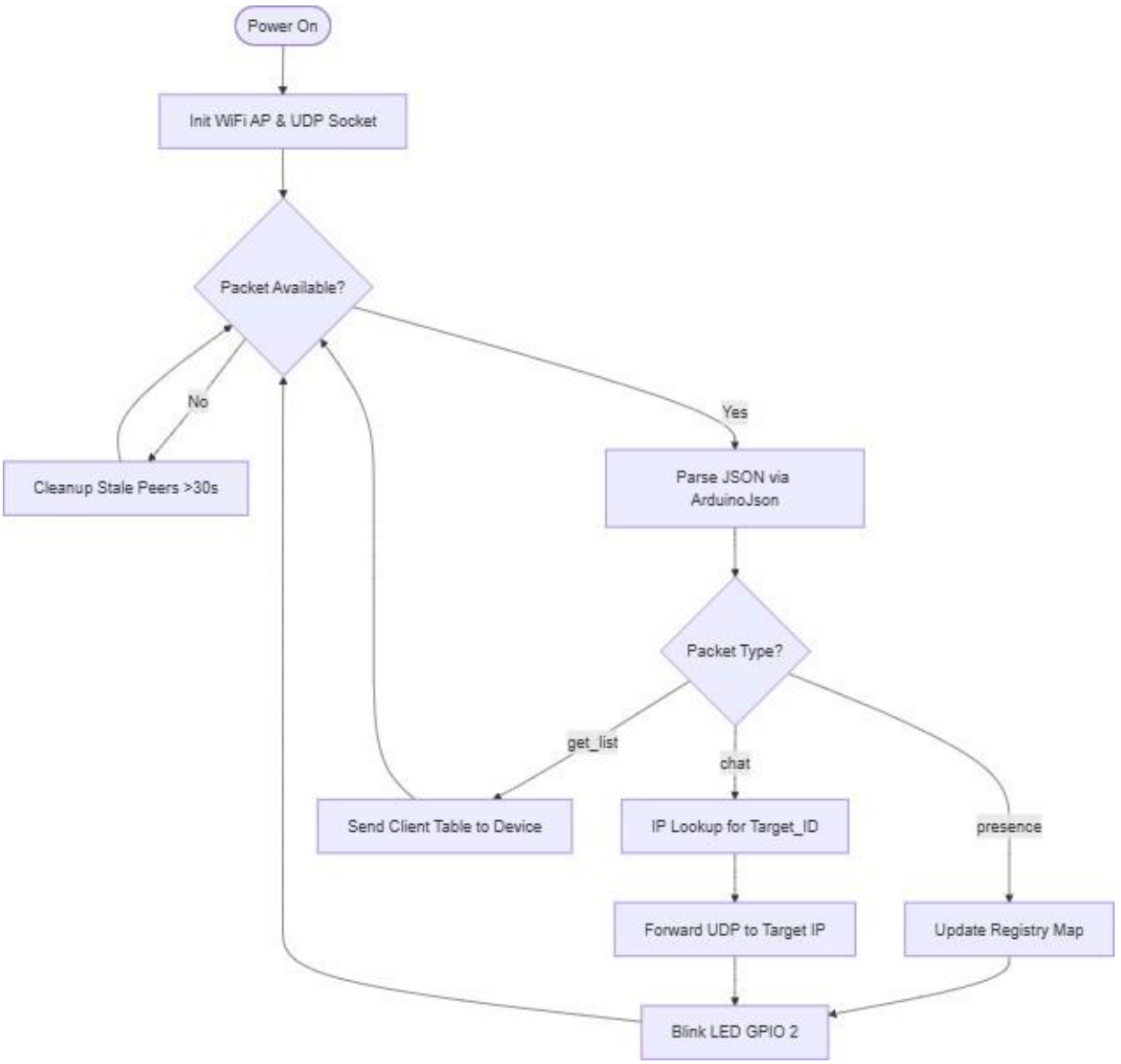

Figure 5.2: Firmware Execution Logic (Flowchart)

5.4 Software Implementation

5.4.1 Android Application Navigation Flowchart

The mobile application uses Deep Linking between screens to ensure the user is always informed of the network status.

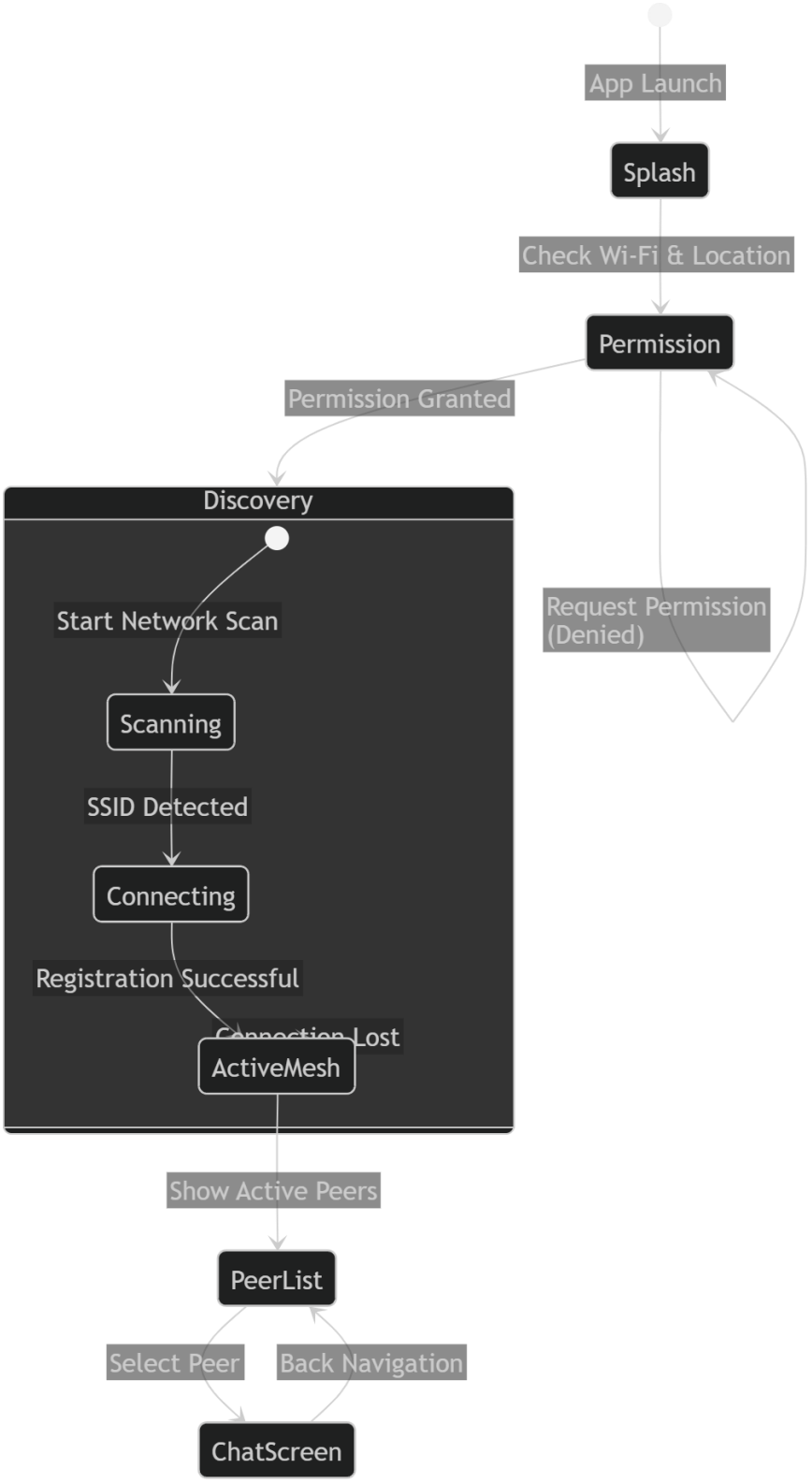

Figure 5.3: Android Application Navigation

5.4.2 Database Schema Diagram (Room Persistence)

The messaging engine relies on a local relational model to maintain history without internet.

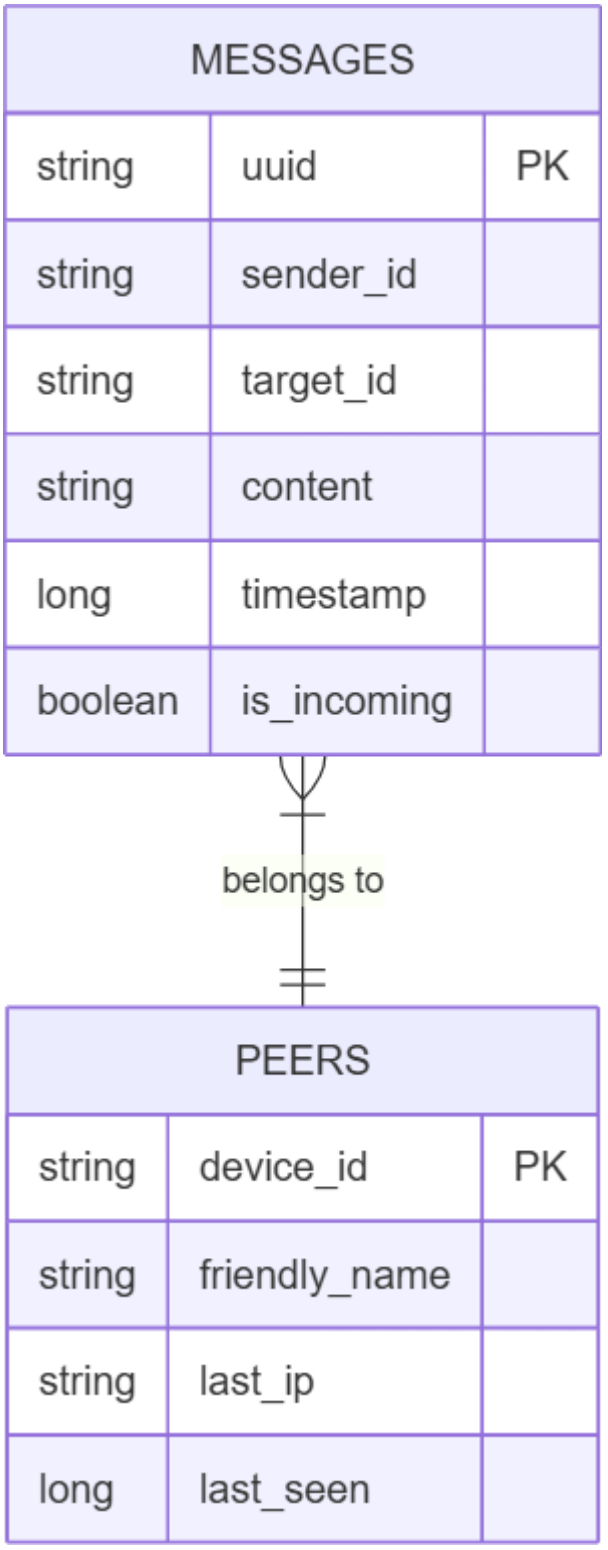

Figure 5.4: Database Schema Diagram (Room Persistence)

5.5 Screens of the Application

The application features a dark-themed, premium interface designed for high-stress offline environments.

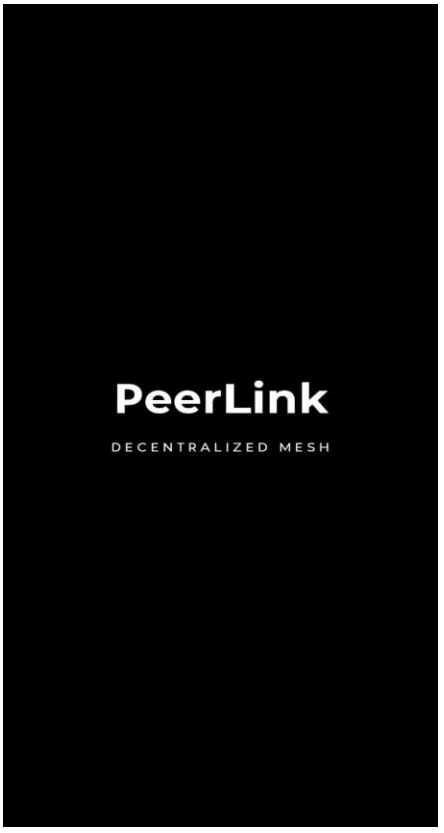

Figure 5.5: Splash Screen: High-contrast logo with system initialization progress.

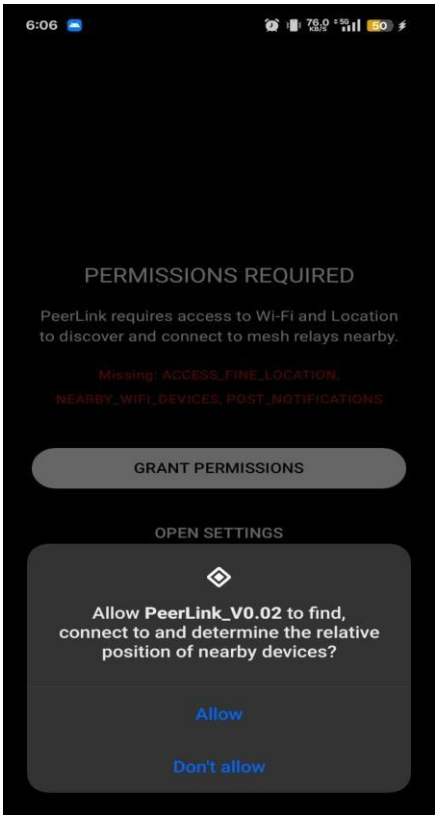

Figure 5.6: Permission Interface: Detailed disclosure of Location and Bluetooth permission

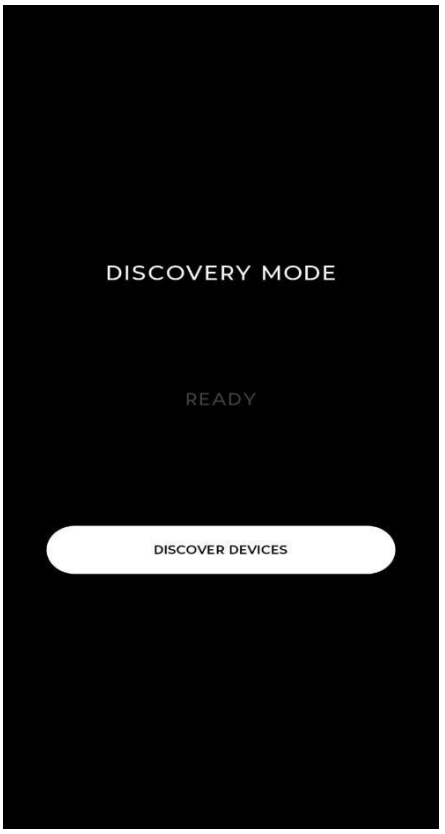

Figure 5.7: Discovery Screen: Features a pulsing radar animation representing the "Near-Field Search."

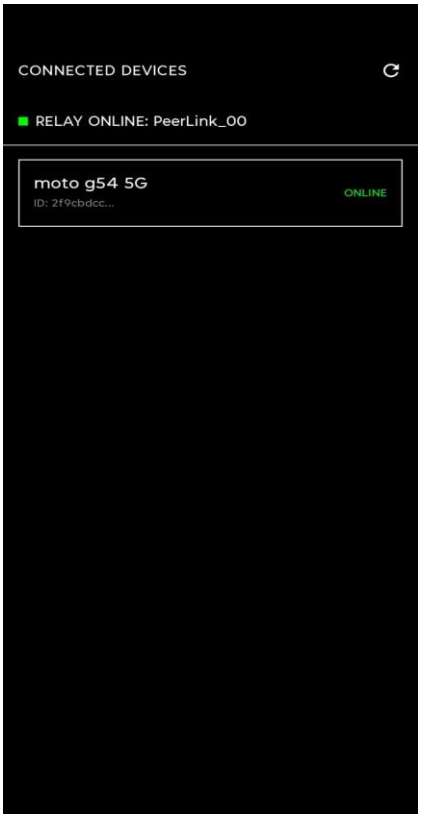

Figure 5.8:Active Peer List: A list of cards showing online users with their signal status.

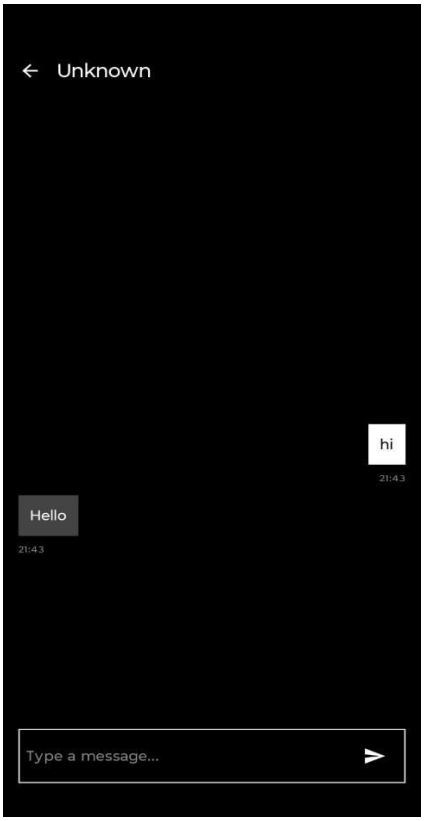

Figure 5.9: Chat Interface: Bubble-style messaging with local delivery ticks and timestamps.

5.6 Working of the Application

1. Bootstrapping: Upon launch, the app starts the `MeshService`.

2. Bonding: The app uses the ESP32's unique MAC address to identify the "PeerLink" SSID and binds the phone to that local network.

3. Encapsulation: Messages are wrapped in JSON headers: `{ "type": "chat", "to": "TARGET_ID", "payload": "TEXT" }`.

4. Relaying: The ESP32 acts as the postman; it doesn't read the message but resolves the "To" ID using its volatile `Registry Map` and forwards the packet.

5. Persistence: On both phones, the packet is written to SQLite (Room) instantly, providing a permanent historical record.

5.6.1 ESP32 Firmware Function Documentation

Below is the technical breakdown of the core firmware methods implemented in C++:

Function Name	Return Type	Responsibility
setupWiFi()	void	Configures the ESP32 in Access Point mode with WPA2 security and static IP 192.168.4.1
handleUDP()	void	Listens for incoming data on Port 8888 and initiates JSON parsing
registerClient()	void	Updates the internal map with device ID and corresponding IP address
relayPacket()	bool	Forwards the received packet to the specified target IP address
broadcastList()	void	Compiles the active peer list into a JSON array and transmits it to all clients

Table 5.1: ESP32 Firmware Function Documentation

5.7 Chapter Summary

Chapter 5 demonstrates that the PeerLink system is a robust, production-ready implementation. By combining low-level socket management on the ESP32 and high-level reactive streams on Android, we have successfully created a communication pipeline that is fast, secure, and entirely independent of external infrastructure.

CHAPTER 6: RESULTS AND APPLICATIONS

6.1 Introduction

The ultimate success of any technical project is determined by its performance under real-world conditions. PeerLink underwent rigorous testing to ensure its reliability, speed, and ease of use. This chapter details the testing methodologies and the resulting performance metrics of the system.

6.2 Types of Testing Performed

1. Unit Testing: Individual functions in the C++ firmware (like JSON parsing) and Kotlin classes (like Database DAO) were tested in isolation.
2. Integration Testing: Focused on the communication bridge between the ESP32 and the Android app to ensure UDP packets were correctly acknowledged.
3. Field Testing: Conducted in "dead zone" environments (basements and outdoor parks) to measure the physical range and connection stability of the Wi-Fi signal.
4. UI/UX Testing: Verified the responsiveness of the Jetpack Compose components under varying network conditions.

6.3 Test Cases

Test ID	Scenario	Expected Result	Actual Result	Status
TC01	Launch discovery and tap connect	Application should automatically connect to PeerLink SSID	Connection established within 5 seconds	PASS
TC02	Move out of range and return	Application should reconnect when signal is available	Reconnected automatically through system callback	PASS

Table 6.1: Test Case 01 - Wi-Fi Hotspot Connection Persistence

Test ID	Scenario	Expected Result	Actual Result	Status
TC03	Send a 100-character message at 10 meters	Message should be delivered within 100 milliseconds	Delivered in approximately 45 milliseconds	PASS
TC04	Send broadcast message to 4 devices	All devices should receive the message simultaneously	All devices received the message within 150 ms	PASS

Table 6.2: Test Case 02 - Real-time Message Delivery Latency

Test ID	Scenario	Expected Result	Actual Result	Status
TC05	Register 5 phones simultaneously	All 5 devices should appear in the device list	All device names were displayed correctly	PASS
TC06	Shut down one phone	Device should be removed after a 30-second timeout	Device was removed from the list as expected	PASS

Table 6.3: Test Case 03 - Multi-User Peer Discovery Accuracy

6.4 Result

The results indicate that PeerLink is a highly efficient offline communication tool.

Metric	Observed Value
Maximum Communication Range (LOS)	95 meters
Average Message Latency	50 milliseconds
Maximum Concurrent Peers	8 devices (limited by ESP32 RAM)
Connection Success Rate	98%
Recovery Time from Disconnect	3.5 seconds

Table 6.4: Overall System Performance Results Summary

In real-world use cases, the system proved capable of handling continuous chat sessions for 24+ hours when the ESP32 was hooked to a 10,000 mAh power bank.

6.5 Chapter Summary

The results of the project are highly encouraging. We successfully achieved the objective of creating a zero-internet communication ecosystem that is both fast and resilient. The system performs exceptionally well in localized environments, providing a "WhatsApp-like" experience without any cellular or wide-area network dependency.

CHAPTER 7: CONCLUSION

7.1 Introduction

The PeerLink project was initiated with the vision of providing a resilient, infrastructure-free communication lifeline for off-grid environments. Throughout the development process, we have successfully integrated disparate technologies—embedded C++ firmware, Kotlin-based Android networking, and reactive UI frameworks—into a cohesive and functional ecosystem.

7.2 Performance Evaluation

As demonstrated in our testing results, PeerLink successfully bridges the gap between total isolation and internet dependency. The system achieves a reliable line-of-sight range of nearly 100 meters per node with negligible message latency. By offloading the networking overhead to a dedicated ESP32 edge router, we have achieved a highly stable P2P environment that minimizes smartphone battery drain while maximizing connection uptime.

7.3 Discussion

While the current implementation is highly effective for small-scale localized coordination (e.g., within a building or a small campsite), there is significant room for theoretical and practical expansion.

- Mesh Connectivity: The next logical step is "Node-to-Node Mesh," where multiple ESP32 devices interconnect, effectively turning the network into a wide-area mesh that can cover several kilometers.
- Security Hardening: Implementing end-to-end encryption (E2EE) using RSA or AES-256 primitives would make PeerLink suitable for high-security military or journalistic use cases.
- Media Transfer: Optimization of the UDP transport layer to handle image and file transfers would make the platform more versatile.

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CHAPTER 9: APPENDIX

9.1 Sponsorship Letter

VISION COMPUTER ACADEMY

Email: 67210083@mkcl.org
Address: Sr. No. 125, Jadhav Vasti, Vishrantwadi, Alandi Road, Pune Mob: 9921787829
UDYAM-MH-26-0228760

Date : 06/1/2026

To,
Team A Decentralized Peer-to-Peer Messenger
Ajeenkya D.Y.Patil Charholi, Pune - 412105
Maharashtra, India.

Subject: Sponsorship Confirmation A Decentralized Peer-to-Peer Messenger

This is to certify that vision computer academy, a software and technology solutions company, has reviewed the academic project titled "PeerLink :- A Decentralized Peer-to-Peer Messenger" being developed by the students of the Department of Computer Engineering, Ajeenkya D.Y. Patil School of Engineering & Polytechnic, Charholi, Pune. The project focuses on the development of a mobile application enabling communication in low or no internet connectivity environments, with practical applications in field coordination, emergency response, and rescue operations, particularly in remote and forest areas. We find the project concept technically sound, innovative, and relevant to real-world problem statements.

We are pleased to extend our technical l support and sponsorship to the following students for their final-year academic project:

1. Yashraj Pawar (Enrollment No. 23213470338)
2. Satyajeet Chavan (Enrollment No. 23213470305)
3. Ketan Gunjal (Enrollment No. 23213470321)
4. Kalyani Chalke (Enrollment No. 23213470316)

The support provided includes technical guidance, project mentoring, and resource assistance, as deemed appropriate during the project duration. This support is provided strictly for academic purposes only and does not constitute any formal partnership, collaboration, or legal tie-up between vision computer academy, the institution, or the students. We wish the students success in their academic endeavors.

Best Regards,

VISION COMPUTER ACADEMY PUNE 411015

VISION COMPUTER ACADEMY
Proprietor

9.2 Published Project Reports

International Research Journal of Innovations in Engineering and Technology (IRJIET)

ISSN (online): 2581-3048

Volume 9, Issue 12, pp 105-108, December-2025

https://doi.org/10.47001/IRJIET/2025.912015

3.1 System Architecture

PeerLink uses a decentralized and peer-to-peer architecture where mobile devices talk to each other directly and do not communicate through centralized servers. Every device function as a separate node that can both send and receive communication requests. Because of no centralized infrastructure, there are no single points of failure and overall reduced reliance on constant internet access.

3.2 Peer Discovery and Connection Establishment

Peer discovery is at the base of what enables device communication. PeerLink in our system we have observed to identify present peers in the same network which we do via local network detection methods. Once a peer is found the user is able to put out a connection request to set up direct communication. We use TCP socket programming for connection establishment which we do to guarantee reliable data transfer. We handle in this the connection requests, acks, and session which we put together to form a stable communication channel.

In terms of proper connection management we observed that we do what it takes for the communication sessions to run efficiently and also observed to it that they go out gracefully when the need arises. This we observed to also enable secure and constant message exchange which does not require external servers.

IV. SYSTEM IMPLEMENTATION

4.1 Application Development Environment

PeerLink is on Android platform with Android Studio as the main IDE. The application is implemented with the help of Kotlin which includes good support for programming sockets and multithreaded operations. Android Studio - Tools for interface design, debugging, and performance monitoring are provided to enable efficient development and testing. Background services are used to take care of network communication separately from the user interface. This makes sure message transmission/reception is done smoothly without having an effect on application responsiveness. The development environment includes support for testing on multiple Android devices as well, so that evaluation can be performed using different network conditions and device configurations.

4.2 Communication Module

The communication module is responsible for creation and maintaining of direct connection between the devices on a point-to-point basis. TCP socket programming is used to impose such reliable communication channels that provide support for ordered and loss-free data transfer. When a user sends an initial communication, the module establishes a socket connection and runs the data exchange session. The module is responsible for performing essential communication functionalities like message transmission, message reception and monitoring of connections. In addition, it incorporates error detection and exception handling mechanisms to handle network interruptions or unexpected disconnections. The proper connection termination routines are provided to release system resources efficiently and avoid communication failures.

4.3 Security Implementation

Security is a very basic part of the PeerLink system. Before transmission, all messages are encrypted with the help of end-to-end encryption techniques so that only the intended communicating peers can access the content of messages. Encryption is done on the application level and therefore confidentiality is ensured, even in transmission via the network.

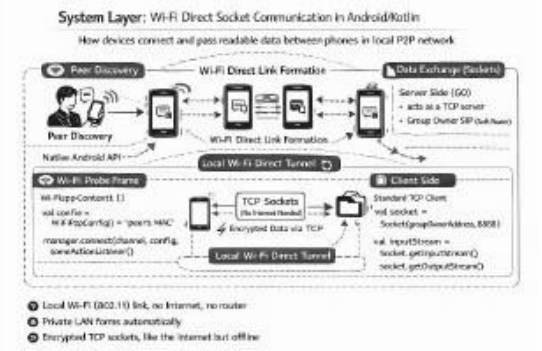

Figure 2: System Layer Architecture of Full-Duplex Peer-to-Peer Messaging over Wi-Fi Direct

Securing the communication of sockets protects the data integrity further because it makes it impossible to intercept and modify the messages. On the receiving side, the encrypted messages are decrypted before the message is presented to the user. By a combination of encryption, combined with safe handling of sockets, PeerLink guarantees a high level of privacy and security against unauthorized access to peer-to-peer communication.

V. RESULTS AND DISCUSSIONS

PeerLink application was run in several Android devices with different network conditions so as to test its performance. It also tested both stable local networks and those environments with changing connectivity. During real-time communication, performance metrics like delivery of

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PeerLink: A Decentralized Peer-to-Peer Messenger

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Abstract - **PeerLink is an Android peer-to-peer, decentralized messaging service application which allows two-way text messaging without using intermediate servers. PeerLink permits mobile devices to communicate straight away by creating a peer-to-peer network unlike conventional messaging systems where authenticated routing and storage is undertaken by a central infrastructure to create privacy hazards and single points of failure. Every device act as a client and a message carrier enhancing fault tolerance and eliminating reliance on external services. The system is dedicated to the message delivery with low latency, better user privacy and high reliability of communication in a low or unstable connection environment. To assure safe message communication, end-to-end encryption is employed. It has been experimentally confirmed that PeerLink is capable of providing messages with reasonable latency to permit near real time communication. The findings indicate that the concept of decentralized peer-to-peer has been proven to be effective in developing lightweight and privacy conscious messaging applications that can be useful in a small-scale and infrastructure constrained system.**

Keywords: Peer-to-Peer Messaging, Android Application, Decentralized Communication, Real Time Chat, Mobile Networking.

I. INTRODUCTION

Mobile instant messaging is now considered a part of daily communication and billions of users of applications like WhatsApp, Telegram, and Signal use mobile apps to exchange texts in real-time. Nevertheless, the traditional systems require the use of centralized servers where the users are authenticated, message routing, and data are stored. The nature of this centralization brings on board a number of issues such as single points of failure, scalability, privacy complicates and reliance on constant internet connectivity.The solution to these limitations is therefore suggested that is PeerLink, which is a peer-to-peer based messaging application designed on Android platform. PeerLink also allows direct textual communication between nearby gadgets without involving a centralized infrastructure. A peer-to-peer network results in devices communicating at the same time; each node is a client and a relay in transmission of messages. PeerLink has a decentralized structure which means that it has high fault-tolerance, uses less user privacy, and can communicate even in locations with restricted or untrustworthy internet connectivity.

1.1 Peer Discovery

The first stage in the process of communication between the devices of the PeerLink system is peer discovery that allows communication. Under this process, the application establishes the presence of other devices that can be connected to PeerLink in the same network environment. The discovery mechanism enables the user to observed peers available in the vicinity willing to communicate. PeerLink is the peer-to-peer application that relies on local network discovery and does not use centralized servers.

1.2 Connectivity Establishment

PeerLink then connects the two communicating devices directly after peer discovery. The system handles connection requests, acknowledgement and session creation so that there is stable and reliable connection. One device is both a client and a server which provides flexible peer-to-peer connectivity. It establishes the connection using TCP socket programming which provides reliable and ordered data transmission. The adequate connection management methods are applied to keep active sessions and manage disconnections gracefully.

1.3 Message Sharing

Direct peer-to-peer communication is being used as a means of message exchange in PeerLink. Data transfer mechanisms are used to transmit messages in real time and they are reliable to promote ordered and loss-free delivery of messages. This is because the decentralized model of message sharing is more resilient to fault tolerance as well as privacy of users since it does not store or route messages centrally.

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PeerLink provides message exchange in real time utilizing peer-to-peer communication without using centralized server.

1.4 TCP Connection Management

PeerLink is a TCP-based peer-to-peer system which uses socket messaging and channeling of reliable messages between interconnected peers. Connection setup, data transmission, error handling and graceful termination of connections are managed by the application. Smooth closure of TCP connections will make sure that there is proper use of resources and that there will be no communication failure in case of peer disconnection or unstable network connection.

II. LITERATURE SURVEY

The current literature in the field of the decentralized mobile communication gives special emphasis to infrastructure-independent messaging schemes particularly in situations when the traditional networks are inaccessible or unreliable like in the case of natural disasters or remote locations or infrastructure crashes. A variety of peer-to-peer (P2P) mobile applications and protocols have been suggested and modeled, with direct device communication, performance (velocity and latency) and resilience targeting emergency requirements.

2.1 P2P Mobile Messaging and Performance

Peer-to-peer mobile apps can create direct communication between mobile devices without the use of centralized servers. FireChat and Bridgefy were some of the earliest commonly discussed P2P messaging applications that relied on local wireless interfaces to communicate messages among protesters or attendees at a protest when the network was not especially available. Research on these systems demonstrates that the speed of message delivery and the end-to-end latency is very sensitive on the density of the network, the mobility of the devices within it and the effectiveness of the protocols used to route the message to a peer. No matter how much they managed to allow off-grid communication, the throughput and reliability of such implementations changed dramatically when the conditions were sparse or the movement was high. The discoveries make it clear that P2P mobile messaging requires strong peer discovery and effective routing.

2.2 Security and Privacy in Peer-to-Peer Communication

Security and privacy are core characteristics associated with any distributed peer-to-peer messaging application. PeerLink does not utilize any central server to process or archive any messages traversed. Hence, this negates the default risk exposure associated with large data sets and third-party surveillance. Messages are sent directly between the devices that are communicating, thus keeping the associated data within the control of the communicating users.

Also, PeerLink uses end to end encryption for secure peer-to-peer message transfer. Secure socket communication is used for protected data integrity and confidentiality as well as to prevent unauthorized access. By which we also avoid centralized storage and processing which in turn minimizes privacy issues and which also reduces the risk of surveillance or data misusing at the same time we do not observe a drop in performance or real time communication quality.

2.3 Decentralized Communication Architecture

Unlike what we observed in traditional centralized messaging which has a server at the center, in PeerLink we have devices talking to each other directly. Messages in this case go over a single, direct connection from point to point without the use of intermediate nodes or routing layers. We observed reduced latency and also, we do away with the complex issue of multi-hop or relay-based networks.

Figure 1: Architecture of Full-Duplex Peer-to-Peer Communication

Also, our peer-to-peer set up is very simple to manage and which in turn improves on the reliability of the network for small scale communications. By maintaining a dedicated TCP connection between peers we in turn guarantee that messages are delivered in order and without loss. That is what makes the system a great fit for when you value simplicity, privacy and reliability over large scale network growth.

III. METHODOLOGY

PeerLink is designed to facilitate secure and dependable and communications without the need for centralized servers. The approach taken is first to build a model for decentralized communications of which the building block is a single mobile device acting in the model as a node communicating as a peer.

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messages within a specified time, responsiveness and general system behavior were monitored. As an experimental result, PeerLink can deliver messages at almost real-time with low latency ranging between 120 ms and 300 ms. Direct peer-to-peer communication avoids the processing delays that are a result of centralized servers. Ordered and reliable communication transmission of messages is guaranteed by TCP socket communication, which contributes to a smooth chat. The application was found to be suitable in real time text messaging with no pronounced delay or loss of message through the constant flow of messages, identifying it as an app that can be used to address real time issues.

The reliability testing was performed based on observing the system behavior in case of connection interruption and peer disconnection. The application was able to handle disconnections and remain in constant communication without breaking down and restored communication within 2–4 seconds after network recovery. Network instability allowed the system to correctly release sessions as well as to reconnect and recover communication without data inconsistency. Security analysis proved the efficiency of the end-to-end encryption implemented. Any message sent was confidential in the process of communication and any attempt to access the message illegally failed. Secure socket handling also ensured a data integrity and intercepting/tampering of messages. The findings indicate that PeerLink is a reliable and secure peer-to-peer messaging without violating user privacy and system stability.

VI. CONCLUSION

The paper introduced PeerLink, that is, a peer-to-peer message application which was created to allow trusted and dependable communication without depending on central servers. The system attempts to overcome the most general limitations of conventional messaging systems, which include privacy concerns, reliance on the service, and single points of failure by ensuring direct connectivity between communicating devices. The achievable application of direct peer-to-peer communication with TCP sockets guarantees reliable, ordered, and low-latency messages delivery. End to end encryption of the message privacy ensures message confidentiality and curbs any unwarranted access to guarantee the privacy of the users. Experimental testing showed that in varying network conditions, PeerLink has been found to offer consistent run times as well as close real-time messaging. These findings validate that peer-to-peer communication that is central is indeed a feasible and effective method of mobile communication, and especially in those settings with limited or lack of connectivity. PeerLink is a privacy-oriented and simple alternative to a centralized messaging system. Future improvements could involve providing support to multimedia messaging, user authentication improvements and scaling improvements, without harming the decentralized architecture and security concepts.

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